FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishmentsof the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts andvalue of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	549024
Award Program	Pollinator Heath Grant
Project Title	Pollinator Health Extension and Metrics in the Pacific Northwest
Grant Start Date	01/01/2019
Grant End Date	12/31/2022
Total Grant Budget	$ 1,091,427.78
Total FFAR Budget	$ 544,928.89
Matching Funder(s)	GloryBee, Central Oregon Seeds, Oregon Department of Agriculture, Oregon State Beekeepers Association
Final Report Date	03/31/2023
Project Director/Principal Investigator Information	
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FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Grantee Organization Information	
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General Information

1.1. Please list the geographic location(s) – city, state, congressional district - wherethe work was conducted. If the work was conducted outside of the US, please list the city and country.
Corvallis, Oregon, Oregon's 4th Congressional District

1.2. How many new jobs were created because of this grant funding?
14

1.3. How many jobs were maintained because of this grant funding?
3

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profitstatus since you were awarded the grant? If yes, please explain.
no

1.5. Has your organization undergone a recent name change? If so, please providethe new name of your organization.
no

2. Scientific Report

2.1. **Public Abstract:** The Pacific Northwest region is home to three managed bee species and up to 900 species of wild bees that contribute to crop pollination. State Legislatures in the region have looked to establish Extension and outreach programs to protect these bees. To meet this demand, our project created comprehensive outreach and educational materials for licensed pesticide applicators and growers to increase the adoption of best management practices to protect bees, resulting in over 9,000 Oregonians and Washingtonians being trained at 84 events. Trainings developed from the grant are now being used across the U.S. We also investigated a key pillar of pesticide training, namely the alignment of hazard warnings on pesticide labels with recommended language from the Environmental Protection Agency (EPA). We discovered that of the 232 labels of pesticides that have a high probability of being exposed to bees, 31.5% deviated from EPA recommendations in at least one way. This finding foregrounded the need for applicators to be taught how to interpret a product label that uses non-EPA-recommended statements to communicate toxicity to honey bees. We also explored the feasibility of monitoring for pesticide residues at a state level to measure the success of educational programs. This involved sampling for pesticides collected by honey bees pollinating four diverse Oregon crops (cherry, clover seed, meadowfoam and carrot seed) and using power analysis to determine the minimum number of fields to detect a 5% difference in the pesticide load. Our research demonstrated that while there is considerable variability among crops, there was too great site-to-site and within site variability to make routine sampling of pollen an economically feasible metric for state-level monitoring. In addition, we released over 200 podcast episodes on Pollinator Health resulting in almost 300,000 downloads by listeners. We also developed the two most successful programs for monitoring for wild bees species in the United States, the PNW Bumble Bee Atlas and the Oregon Bee Atlas. The two programs have resulted in over 400 people in the region trained in bee survey, over 140,000 new bee occurrence records and the largest contemporary bee-plant network in the world. Both programs have been adopted across the U.S. and are the model for surveying bumble bees and other wild bee species.

2.2. **A description or interpretation of how the key findings of the project can be used by the agriculture community.** The project provides guidance and models for states across the U.S. protect managed and wild bees, ensuring the availability of bee stocks and populations for crop pollination. Our approach to developing Bee Protection Protocols brought together crop consultants, growers, beekeepers and Extension personal for regular meetings to develop an agreement on how to keep bees protected during pollination. This approach resulted in a large increase in the number of growers adopting practices associated with reducing exposure of bees to pesticides and is a model for other states looking to increase productive communication among growers, pesticide applicators and beekeepers. The grant also resulted in five new Extension products to help growers better understand how to use pesticides safely around bees, with over 9,000 licensed pesticide applicators trained in Washington and Oregon. Two of these products, a guide to understanding the hazards indicated on pesticide labels and a training on pesticide label comprehension, have been adopted across the U.S. Our research has also identified widespread problems associated with information on bee hazards conveyed on pesticide labels, which may help improve the labels to better protect honey bee colonies. The project helped launch and grow a podcast on pollinator Health, PolliNation, which has not only become one of the largest shows of its kind in the U.S., but has resulted in an improved perception of the role of agriculture in improving bee health.

2.3. **Research Outcomes/Impact.** The primary goal here is toanswer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that couldimpact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices,legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations havecontributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Goal 1. Pollinator Health Extension and Outreach.
Pesticide Training. The project has made significant inroads in increasing pesticide applicator comprehension on hazards of pesticides to bees, with over 9,000 license applicator trained across 84 events. To assess whether these extension and outreach efforts were reducing pesticide poisoning incidents, we surveyed commercial Oregon beekeepers in July 2021. They indicated that since the onset of our education and outreach campaign, they have experienced a large to moderately large reduction in pesticide exposure to bees across several crops that were specifically targeted by our training (n=19). This decrease in the level of pesticide poisoning incidence will help ensure the sustainability of paid and unpaid pollinator services back to more than 15 different crops. Being able to document effective education may have broader impacts for Oregon agriculture. For example, it may help applicators retain access to pesticides that have been or are in the process of being restricted owing to pollinator concern. In the Province of Ontario, Canada, for example, restrictions around pesticide use owing to concerns around honey bee exposure, has reduced farm revenues by an estimated $630 million annually.

Pesticide Label Analysis. We also conducted an analysis of hazard language on 232 pesticide labels used in Oregon and Washington around bees and found incongruencies from the Environmental Protection Agency’s (EPA) guidelines, which was the first time the extent of problems with the hazards to bees communicated on pesticide labels was documented. This study has had the short-term impact of helping pesticide educators identify pesticide products with problematic language, which they are now able to convey to their students. More long term, the identification of patterns of label incongruencies will help EPA standardize these labels as the products come up for their scheduled reviews, resulting in safer use of products by pesticide applicators.

PolliNation Podcast. The podcast has had the immediate effect of increasing awareness of bee protection initiatives in the region and has become established medium for communicating pollinator health initiatives across the Pacific Northwest in the long term.

Goal 2. Metrics for Honey Bee Pesticide Exposure and Pollen Nutrition
Analysis of trapped honey bee pollen for pesticide residues is one way of monitoring for changes in pesticide hazard across a landscape and over a bloom period. Our results cast doubt on the use of pollen sampling to detect landscape changes in pesticide hazard to honey bees, resulting in the short term impact of discouraging states from using this technique to monitor for reduced pesticide exposure to bees, and more long term, looking for more suitable methods to survey for exposure. We also conducted a detailed analysis of pesticide exposure in sweet cherry product areas, comparing plant foraging preference to their pesticide exposure for managed honey bees and wild bees. Our study demonstrated that managed and wild bees have very different foraging preferences, and in turn, pesticide exposure patterns. This study puts pressure on assumptions of the Environmental Protection Agency's risk assessment framework for bees, which assumes that exposure to bees is driven by the attractiveness of plants to honey bees.

Goal 3. PNW Bee Atlas
The Atlas has trained over 220 citizen scientists, organized into 14-regionally-based teams around Oregon. The program has been exceptionally popular with the public and was featured on Oregon Public Broadcasting's program Oregon Field Guide in 2018 (*The quest to find every kind of bee in Oregon*). Nevertheless, the Atlas faced issues similar to other high-level community science programs. Despite our exceptionally high data quality, we had a high number of 'dabblers' in 2018 who only episodically collected, producing few

specimens, virtually all of which were common species. This phenomenon has been observed in many birding-based citizen science projects. By modifying and intensifying our training in 2019 and transitioning the volunteer program to a Master Certificate program in 2020 (the Master Melittologists) we have doubled the number of volunteers submitting larger collections and have increased the scientific significance of the bees collected. Volunteers collected 136,220 new bee specimens from every county in Oregon from 1,456 unique plant genera making this the largest contemporary bee plant network in the U.S. To date, 400 species have been named using traditional microscopy, 125 species using DNA barcoding methods and an estimated 150 morophogroups, resulting in an estimated 625 species detected, which is
approximately 80% of the known species known to the state. New records include dozens of species whose presence in Oregon had not been recorded for decades. Remarkably, most of these specimens were captured when the bee was visiting flowers, and volunteers have generated a database of photo-vouchers of these host plants, enabling an unprecedented view of the plant-pollinator networks on the scale of a state. This makes the Atlas not only the largest volunteer native bee inventory in the US, but also, the largest effort to document bee-plant interactions anywhere in North America. Finally, the project contributed to the PNW Bumble Bee Atlas, which generated an additional 40,434 bumble bee records, consisting of 25 species, for Oregon, Washington and Idaho. The immediate impact of this work was to create the framework for bee survey in the U.S. The Master Melittologist program now trains volunteers in specimen-based collection surveys in Oregon, Washington, Idaho and British Columbia, Canada. The Bumble Bee Atlas has expanded to Nebraska, Minnesota, California, Missouri and Iowa.

2.4. What were the goals/specific aims of the project? If the approved application listsmilestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Goal 1. Pollinator Health Extension and Outreach.

- Develop Best Management Practices for managed pollinators and track grower adoption.
- Train license pesticide applicators on provisions on pesticide labels designed to protect bees and track training.
- Produce a podcast designed to promote pollinator health practices.

Goal 2. Metrics for Honey Bee Pesticide Exposure and Pollen Nutrition

- Collect pollen from commercial cropping systems in Oregon to determine the floral composition of pollen collected and whether pollen sampling could be used to measure pesticide exposure on a state level.

paper published with two additional manuscripts in preparation (100% complete).

Goal 3. PNW Bee Atlas

- Create a bumble bee and wild bee atlas for the state of Oregon and digitize historic specimens of Bombus spp. and Megachilidae species.

2.5. Have any of the major goals/specific aims or milestones for the project changedsince the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (bothpositive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Goal 1. Pollinator Health Extension and Outreach.

- The Clover Seed Bee Protection Protocol was developed, and the Vegetable Seed Bee Protection Protocol was revised following meetings involving the Extension and Research Team, the Advisory Committee, representatives from the Oregon Clover Commission, Specialty Seed Growers of Western Oregon and Oregon State Beekeepers Association and selected crop consultants. The Clover Seed Bee Protection Protocol was developed through two meetings in January 2020. The revision of the Vegetable Seed Bee Protection Protocol was completed through a single meeting in February 2020. We used electronic clickers at the annual meetings of the Oregon Clover Commission and Specialty Seed Growers of Western Oregon in winter 2020 to evaluate the standard industry practices to help inform the Protocols. Card form summaries of the Protocols were provided to growers in 2020, with a more extensive guide published in 2022 (Vegetable Seed) and 2023 (Clover Seed). BMPs complete (Bee Protection Protocols) and published as guides and as infographic cards to use at trainings. and grower knowledge survey completed. When growers were asked "how many people do you estimate you will share some aspect of this project within the next 12 months?" clover seed growers (n=91) and vegetable seed growers (n=10) indicated, on average, would share the project with 3.9 and 6.3 additional growers, respectively. Moreover, all growers indicated that the project "improved average awareness" of bee health and the large majority indicated the project had changed their attitudes to bee health and that they were likely to "adopt one or more of the practices" outlined in the Bee Protection Protocol.
- A training was developed for pesticide applicators that focused on understanding aspects of the pesticide label indicating hazards and risks to bees. The training was adapted to an online training for self-paced education (https://workspace.oregonstate.edu/course/getting-tough-with-pests-and-going-soft-on-pollinators, adapted for a national audience in 2023 - https://www.canr.msu.edu/courses/pollinator-protection) A total of 9,000 applicators were trained in pesticide label comprehension, and we have documented improvements in pesticide label comprehension. Data collected during trainings indicate that approximately 25% of applicators understood hazard statements on pesticides labels, and this was increased to 96% after training.

- We conducted research that analyzed the EPA master labels of insecticides used in 16 situations where there is concern over Oregon honey bee colony exposure to pesticides. Of the 232 labels analyzed, 31.5% deviated from EPA recommendations in at least one way. Labels of both commercial and garden products deviated from USEPA recommendations. Products with labels that deviated from EPA recommendations were found across application situation and active ingredient subgroup. The percentage of products per application situation and per chemical subgroup whose label deviated from USEPA recommendations varied. The most common deviation represented a direct contradiction between the statement used and the toxicity class of the product's active ingredient.
- The PolliNation podcast has become a resource for pollinator health nationally. Episodes of the show have been downloaded 292,067 times (as of Dec 31, 2022) with approximately 20% of the downloads within Oregon and 80% within the US. PolliNation currently has listeners in every U.S. state and Canadian province, as well as listeners in 130 additional countries. Moreover, we use excerpts from the podcast in trainings as means of showcasing the kind of innovation that is taking place in the state. To date (Dec 31, 2022) 72 listeners have responded to the PolliNation podcast listener survey. The survey uses a Likert scale to determine the degree of listener agreement with statements about the podcast (1=strong agreement and 5=strong disagreement). Listeners have indicated that they agree (average score = 2.2) that they have adopted tips featured in the episode. Listeners agreed that the podcast "improved my confidence in efforts to protect pollinators in Oregon" (average score = 1.98). They also responded that the podcast influenced their perception of the position of Oregon's pollinator protection efforts relative to other states ("Oregon leads efforts to improve state-level pollinator health in the US," - average score = 1.80). They also stated that the podcast enhanced their perception that "efforts [are] being taken by commercial land managers to protect pollinators" (average score = 1.92).

Goal 2. Metrics for Honey Bee Pesticide Exposure and Pollen Nutrition

- Analysis of trapped honey bee pollen for pesticide residues is one way of monitoring for changes in pesticide hazard across a landscape and over a bloom period. We collected and analyzed pollen from 70 sites over two years in four different cropping systems: carrot seed, sweet cherry, clover seed, and meadowfoam in Oregon state. Each pollen sample was tested for 245 pesticide residues as a whole sample; next, each pollen sample was sorted into its component parts and retested for pesticide residues. We then used power analysis to determine how many sites would be needed to clearly resolve differences in Hazard Quotient (HQ) at 50, 100, and 1000 threshold levels. Our analysis found that the variation in HQ was too high to use pesticide residue sampling as a cost-effective monitoring technique. Sweet cherry had statistically higher HQ values than each other crop. However, no significant differences were found between focal crop pollen (i.e. cherry pollen within a cherry system) and non-focal crop pollen (i.e. all other pollen types found within a cherry system). As indicated in the section above, the results cast doubt on the use of pollen sampling to detect landscape changes in pesticide hazard to honey bees.

- A more detailed analysis of pesticide risks was conducted in sweet cherry. The study tested the hypotheses that pesticide risks to different bee taxa is equal and that land use patterns do not influence risks. The study was repeated across two years in the main commercial sweet cherry producing region in Oregon, the Dalles. Each year pollen was trapped from 12 locations from commercial pollination honey bee hives (24 samples) and collecting pollen from 18 hives established by the PI throughout the peak of cover crop bloom at five different times (90 samples). Furthermore, mason bee nesting sites were established at 14 sites and collected, and processed at the end of the season. Finally, all wild bees were collected from orchards and surrounding native plant community and from cover crop fields, and the species of plants the bees were visiting was recorded. Pesticide testing was conducted on pollen collected from bees, both as a composite sample, but also to determine pesticide levels in each species of plant represented in the pollen. In order to get the later values, pollen samples were divided into pollen morpho-groups (i.e., pollen with a similar color pattern). Each pollen morpho-group underwent acteolysis to allow identification of the pollen. Finally, morpho-group is tested for pesticides and compared with the composite pollen pesticide sample. We found that while pesticide risk was highest in pollen collected by honey bees, this was from cherry tree. Notably, the pesticide risk was lowest among wild bees, which were rarely seen visiting cherry.

Goal 3. PNW Bee Atlas

- Pacific Northwest Bumble Bee Atlas. A total of 276 volunteers were recruited and trained and contributed 40,434 bumble bee records. The bumble bee records represent 25 different species. Volunteers were trained in person or though educational material provided through a website (www.PNWBumbleBeeAtlas.org). The work has resulted in five journal articles and two reports.
- We digitized all historic Bombus spp. and Megachilidae specimens from Oregon State
  Arthropod collection resulting in 9,550 historical specimens being fully captured.
- Interactive Bee Atlas: Website launched and data from 2018 and 2019 loaded up (https://oregonbees.osac.oregonstate.edu).
- Oregon Bee Atlas: Milestones -> A total of 221 volunteers were recruited and trained using an combination of self-paced online training (6 module online training program) and in-person training including an annual five day bee taxonomy course. Volunteers completing the training were provided a certificate, as an Apprentice-Level Master Melittologist (the first program of its kind in the U.S.).

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Research questions were addressed using testable hypothesis and treatments were assigned using randomization. In survey or descriptive aspects of the work, we collected data anonymously and used categories that avoided a-priori interpretation. Work involving taxonomy of bees, vouchers were deposited at museums (Oregon Bee Atlas) or made available online (PNW Bumble Bee Atlas), making it possible for other taxonomists to confirm the determinations.

2.8. List key stakeholders that could be served by results of this project.

Commercial beekeepers involved in crop pollination.
Growers of crops that use pollination.
Licensed pesticide applicators.
State and federal agencies charged with the protection of rare, threatened or endangered animals.
Members of the public invested in the protection of bees (e.g., gardeners, Extension volunteers, conservationists).

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sourceswill be reported under Information Products. (up to 1,000-word limit)

Extension presentations (selection, more available upon request):

2022

Pesticides Trainings:

Getting tough with pests and going soft on pollinators:

- Washington State University Pesticide Safety Education Program – End of Year (webinar), WA, December 8, 2021, (343 participants)
- Oregon Agriculture Chemicals & Fertilizer Association (webinar), OR, October 11, 2022, (101)
- Oregon State University Pesticide Safety Education Program Non-Ag Core (webinar), OR, October 11, 2022, (100)
- Nut Grower Society, Albany, OR, August 3, 2022, (84)

- Young Harris Beekeeping Institute, Young Harris, GA, May 20, 2022, (127)

Bee Protection Protocols:
- Alberta Beekeeper Commission, Edmonton, AB, Canada, December 3, 2022, (210)
- Southern Alberta Beekeeping Association Conference, Lethbridge, AB, Canada, June 16, 2022, (83)
- Oregon Clover Commission Meeting, Wilsonville, OR, February 2, 2022, (237)
- Specialty Seed Growers of Western Oregon Meeting (webinar), January 20, 2022, (23)

Bee Atlas:
- Winter Dreams Summer Gardens Symposium (webinar), Josephine County Master Gardeners, OR, November 3, 2022, (58)
- Lane County Beekeepers Association, Eugene, OR, October 23, 2022, (36)
- University of Lethbridge Biology Lecture Series, Lethbridge, AB, Canada, June 15, 2022, (32)
- Young Harris Beekeeping Institute, Young Harris, GA, May 20, 2022, (34)
- NW Natural Education Series (webinar), OR, February 24, 2022, (39)
- Young Harris Beekeeping Institute, Young Harris, GA, May 21, 2022, (127)

<u>*2021*</u>
Pesticides Trainings:
- IPM4Bees, North-Central IPM Center (webinar), December 1, 2021 (35)
  National
- League of Woman Voters (webinar), October 12, 2021 (42)
- Portland Urban Beekeepers (webinar), October 6, 2021 (30)

- Oregonians for Food and Shelter Pesticide Training Session, Albany, OR, November 18, 2021 (183)
- Oregon Agriculture Chemicals & Fertilizer Association Training (online), November 4, 2021, (275)
- CORE Training, Pesticide Safety Education Program (online), October 20, 2021 (125)

- North American Raspberry and Blackberry Conference (webinar), February 25, 2021 (35) **International**
- Lower Mainland Horticultural Improvement Association, Horticultural Growers Short Course (webinar), February 25, 2021, (78) **International**
- Chemical Applicator Course, January 19, 2021, (100)
- Pesticide Safety Education Program, Chemical Applicator Course (webinar), January 6, 2021, (100)

Bee Protection Protocols:
- Specialty Seed Growers of Western Oregon (webinar), January 19, 2021, (18)

Bee Atlas:
- Agriculture and Climate Change Resilience in the National Scenic Area (webinar) November 18, 2021 (42)
- Rotary Club of Eugene-Delta (webinar) November 12, 2021 (38)
- Colorado Pollinator Summit (webinar), November 4, 2021 (123) **National**
- Oregon State Beekeepers Association, Florence, OR, October 23, 2021 (115)
- Multnomah Master Gardeners (webinar), July 16, 2021 (87)
- Central Oregon Beekeepers Association (webinar), June 30, 2021 (37)
- Deschutes Land Trust Nature Nights (webinar), May 20, 2021 (38)
- Oregon State University Science on Tap (webinar), May 6, 2021 (73)38)
- Johnson Creek Watershed (webinar), March 18, 2021 (45)
- Linn Benton Beekeepers Association (webinar), February 17, 2021 (28)

2020
Pesticide Trainings:
- Pesticide Safety Education Program, Agriculture CORE Course (webinar), OR, December 5, 2020 (69)
- Willamette Valley Ag Expo (webinar), OR, November 19, 2020 (221)
- Pesticide Safety Education Program, Willamette Valley Course (webinar), OR, November 19, 2020 (128)
- Willamette Valley Ag Expo (webinar), OR, November 17, 2020 (225)

- Commercial Pesticide Applicator Training Program (webinar), KS, November 13, 2020 (80) **National**
- Pesticide Safety Seminar, Oregon Agriculture Chemical and Fertilizers Association (webinar), November 10, 2020 (259)
- Tualatin Valley Beekeepers Association (webinar), OR, June 30, 2020, (25)
- Pesticide Recertification Short Course, La Grande, OR, March 5, 2020, (52)
- Urban Pesticide Recertification Course, Oregon City, OR, February 19, 2020, (95)
- Malheur County Pest Management Course, Ontario, OR, February 18, 2020, (124)
- Pesticide Safety Education Program, Non Crop Vegetation Management Course, Albany, OR, January 30, 2020, (225)
- Pesticide Recertification Short Course, Central Point, OR, January 28, 2020, (117)
- Pesticide Safety Education Program, Chemical Applicator Short Course, Wilsonville, OR, January 7, 2020, (155)

Bee Protection Protocols:

- Oregon Clover Commission, Wilsonville, OR, February 5, 2020, (152)

Bee Atlas:

- Oregon State Beekeepers Association, November 4, 2020, (97)
- Portland Urban Beekeepers, October 7, 2020, (28)

2019

Pesticides Trainings:

- Pesticide Safety Education Program, CORE in the Gorge, Hood River, OR, December 10, 2019, (52)
- Northwest Landscape Expo, Oregon Landscape Contractors Association, Portand, OR, November 19, 2019, (52)
- Pesticide Safety Education Program, Willamette Valley CORE Course, Corvallis, OR, November 19, 2019, (165)
- Pesticide Safety Education Program, North Coast Non Crop Course, Astoria, OR, November 13, 2019, (62)
- Oregon Association of Nurseries, Aurora, OR, November 12, 2019, (12)

- Oregon Agricultural Chemicals & Fertilizers Association (OACFA) Safety and Stewardship Seminar, Wilsonville, OR, November 7, 2017, (229)
- Pesticide Safety Education Program, Forestry & Natural Resources Course, Roseburg, OR, November 6, 2019, (56)
- Oregon Agricultural Chemicals & Fertilizers Association (OACFA) Safety and Stewardship Seminar, Springfield, OR, November 6, 2018, (225)
- Oregon Agricultural Chemicals & Fertilizers Association (OACFA) Safety and Stewardship Seminar, Pendleton, OR, November 5, 2019, (175)
- Pesticide Safety Education Program, Statewide CORE Video Conference, October 3, 2019, (85)
- Oregon Bee Project Flagship Farm Webinar, August 29, 2019, (5)
- Washington State University Cranberry Field Day, Long Beach, WA, July 31, 2019, (62) **National**
- Southern Alberta Beekeepers Association, Brooks, AB, Canada, March 8, 2019, (83) **International**
- Oregon Clover Commission Annual Meeting, Wilsonville, OR, February 6, 2019, (125)
- Cranberry School, Bandon, OR, January 31, 2019, (65)
- Pesticide Applicator Recertification, Central Point, OR, January 28, 2019, (127)

Bee Atlas:

- Lewis and Clark National Historic Park, Astoria, OR, November 13, 2019 (32)
- Oregon Master Naturalist Conference, Corvallis, OR, October 5, 2019 (46)

Presentations to peers (selection, more available upon request):

Melathopoulos, A. P., Best, L., Feuerborn, C., Fischer, H. A., Kincaid, S., Holt, J. and Marshall, C. PNW Cit Sci Summit (virtual), "The Oregon Bee Atlas: Statewide inventory of native bees with volunteer Master Melittologists," (October 4, 2022).

Melathopoulos, A. P., Ates, S., Moretti, M., DeBano, S., and Sagili, R. Western IPM Center IPM Hour, "IPM + Bees? Perfect match or "swipe left"," (March 1, 2022).

Melathopoulos, A. P., Anderson, N. P., Buckland, K., Pacific Branch of Entomology Society of America Annual Meeting (virtual), "Getting Real: Increased Adoption of Bee-Friendly Farming through Industry-Led Bee Protection Protocols in Oregon. Symposium: Supporting Pollinators and Pollination in the PNW," Virtual. Regional, Invited. (April 6, 2021).
Melathopoulos, A. P., Best, L., Feuerborn, C., Fischer, H. A., Kincaid, S., Holt, J. and Marshall, C. Association of Natural Resource Extension Professionals Annual Meeting (virtual), "The Oregon Bee Atlas: Statewide inventory of native bees with volunteer Master Melittologists," (May 25, 2021).
Melathopoulos, A. P., and Best, L. R. US National Native Bee Monitoring Research Coordination Network (RCN) Meeting (virtual), "Oregon Bee Atlas: Statewide inventory of native bees with volunteer Master Melittologists," (May 13, 2021). Invited
Melathopoulos, A. P., Anderson, N. and Buckland, K. Symposium: Supporting Pollinators and Pollination in the PNW, Entomological Society of America, Pacific Branch Meeting Pacific Branch (virtual), "Getting real: Increased adoption of bee-friendly farming through industry-led Bee Protection Protocols in Oregon," (April 6, 2021). Invited
Melathopoulos, A. P., Best, L., Feuerborn, C., Fischer, H. A., Kincaid, S., Holt, J. and Marshall, C. Surveying for Native Bees on Federally Managed Land, National Council for Air and Stream Improvement (NCASI) (virtual), "The Oregon Bee Atlas: Statewide inventory of native bees with volunteer Master Melittologists," (January 19, 2021).
Melathopoulos, A. P., Best, L., Feuerborn, C., Fischer, H. A., Kincaid, S., Holt, J. and Marshall, C. Entomological Society of America Annual Meeting (virtual), "The Oregon Bee Atlas: Statewide inventory of native bees with volunteer Master Melittologists," (November 12, 2020).
Melathopoulos, A. P., Agricultural Stewardship and Best Management Practices to Reduce Pollinator Risk, Environmental Protection Agency (EPA) Webinars on Pollinator Health and Habitat (webinar), "The Oregon Bee Project: Case study of best management practices in a specialty crop state," (August 18, 2020).
Melathopoulos, A. P., Western Region Honey Bee Research Seminar, "Smoothing out the rough spots: Bee protection protocols for pollination in Oregon," (webinar) (June 29, 2020).

2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

We had significant difficulties in recruiting graduate students, which resulted in two failed searches. This resulted in us requesting a no-cost extension to the project. Furthermore, because of the late recruitment of a PhD student for Goal 2, the project was initiated just as COVID-19 lockdown was initiated. We were, however, able to proceed with the work, but with reduced staff, resulting in the PI devoting significantly more time to assisting the PhD student in the field than anticipated.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to providetraining and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to500-word limit)

Nothing to report.

2.12. Please indicate the number of undergraduate and graduate students, post- doctoral scholars, or other educational components involved during this reportingperiod. If other education components are involved, please describe them in detail. (up to 500-word limit)

The project involved one PhD student and 16 undergraduates. Notably, two of the undergraduates conducted peer-reviewed research and earned a publication because of this project. In addition, the project also trained an estimated of 9,500 growers and pesticide applicators and approximately 500 citizen scientists in wild bee survey techniques.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video,audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

- Reports or monographs: 2
- Extension products:8
- Scholarly publications: 9 published, 3 pending
- Websites: 3
- Training platforms: 2

3.2. Please provide a list of citations for the information products produced during theproject.

Reports or monographs:

Hatfield, R., K. Merg, and A. J. Sauder. (**2021**). A PNWBBA Guide to Habitat Management for Bumble Bees in the Pacific Northwest (No. 21-024). The Xerces Society for Invertebrate Conservation.

Hatfield, R., L. Svancara, L. Richardson, J. Sauder, and A. A. Potter. (**2021**). The Pacific Northwest Bumble Bee Atlas: Summary and Species Accounts (No. 21-026). The Xerces Society for Invertebrate Conservation.

Extension products:

Melathopoulos, A., Anderson, N. and Sagili, R. *Submitted.* *Bee Protection Protocol for Clover Seed Production*. Corvallis, OR: OSU Extension.

Kaur, N. (2023). *2023 PNW Insect Management Handbook*. Corvallis, OR: OSU Extension.

NOTE: Complete revision of the bee protection sections of the Handbook.

Melathopoulos, A., Buckland, K. and Sagili, R. (2022). *Bee Protection Protocol for Western Specialty Seed Crops*. EM 9353, Corvallis, OR: OSU Extension. https://extension.oregonstate.edu/pub/em-9353

Melathopoulos, A., Bell, N., Danler, S., Detweiler, A.J., Kormann, I., Langellotto-Rhodaback, G.A, Sanchez, N., Smitley, D. (2020). *Enhancing Urban and Suburban Landscapes to Protect Pollinators* (pp. 41). EM 9289, Corvallis, OR: OSU Extension. https://catalog.extension.oregonstate.edu/em9289

Melathopoulos, A. P., Kormann, I., Anderson, N. (2020) Bee Protection Protocol for Clover Seed.

Melathopoulos, A. P., Kormann, I., Anderson, N. (2019) Bee Protection Protocol for Vegetable Seed.

Melathopoulos, A. P., Kormann, I., Kachadoorian, R., Uribe, G. (2019) Protect bees: Read Pesticide Labels. [adapted for training in California by California Almond Board, nationally through North American Pollinator Protection Campaign]

Melathopoulos, A. P., Kormann, I., Kachadoorian, R., Uribe, G. (2019) Proteja a las abejas: Lea las etiquetas de los pesticidas.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during theproject resulting directly from the FFAR award? If yes, please provide citation.

Swanson, L., Bucy, M. and Melathopoulos, A., *Submitted.* Residual toxicity of pesticides to bees and correspondence to information on pesticide labels using data from published studies. PeerJ.

Carlson, E.A., Melathopoulos, A. and Sagili, R., *Submitted.* There is insufficient power to detect changes in pesticide hazard to honey bees from trapped pollen: A study across four cropping systems. PeerJ.

Colgan, A. M., R. G. Hatfield, A. Dolan, W. Velman, R. E. Newton, and T. A. Graves. *In Review.* Quantifying Effectiveness and Best Practices for Bumble Bee Identification from Photographs. Sci. Rep.

Janousek, W. M., M. R. Douglas, S. Cannings, M. A. Clément, C. M. Delphia, J. G. Everett, R. G. Hatfield, D. A. Keinath, J. B. U. Koch, L. M. McCabe, J. M. Mola, J. E. Ogilvie, I. Rangwala, L. L. Richardson, A. T. Rohde, J. P. Strange, L. M. Tronstad, and T. A. Graves. (2023). Recent and future declines of a historically widespread pollinator linked to climate, land cover, and pesticides. Proceedings of the National Academy of Sciences. 120: e2211223120.

Carlson, E.A., Melathopoulos, A. and Sagili, R., (2022). The value of hazard quotients in honey bee (*Apis mellifera*) ecotoxicology: A review. Frontiers in Ecology and Evolution. https://doi.org/10.3389/fevo.2022.824992

Best, L., Engler, J., Feuerborn, C., Larsen, J., Lindh, B., Marshall, C.J., Melathopoulos, A., Kincaid, S. and Robinson, S.V., (2022). Oregon Bee Atlas: wild bee findings from 2019. Catalog: Oregon State Arthropod Collection, 6(1). 10.5399/osu/cat_osac.6.1.4906

Best, L., Engler, J., Feuerborn, C., Larsen, J., Lindh, B., Kincaid, S., Melathopoulos, A., Marshall, C. J. (2022). Oregon Bee Atlas Survey Data: 2019. Version 1.3. Oregon State University. Occurrence dataset https://doi.org/10.15468/kuwm6h accessed via GBIF.org on 2023-01-30.

Best, L., Marshall, C., Feuerborn, C., Holt, J., Kincaid, S., Melathopoulos, A.P. and Robinson, S. (2021). Oregon Bee Atlas: native bee findings from 2018. Catalog: Oregon State Arthropod Collection. 5. 10.5399/osu/cat_osac.5.1.4647.

Best, L., Feuerborn, C., Holt, J., Kincaid, S., Marshall C. and Melathopolous A. (2021). Oregon Bee Atlas Survey Data: 2018. Version 1.4. Oregon State University. Occurrence dataset https://doi.org/10.15468/dmyc73 accessed via GBIF.org on 2021-03-03.

Graves, T. A., W. M. Janousek, S. M. Gaulke, A. C. Nicholas, D. A. Keinath, C. M. Bell, S. Cannings, R. G. Hatfield, J. M. Heron, J. B. Koch, H. L. Loffland, L. L. Richardson, A. T. Rohde, J. Rykken, J. P. Strange, L. M. Tronstad, and C. S. Sheffield. (2020). Western bumble bee: declines in the continental United States and range-wide information gaps. Ecosphere. 11: 99.

Bucy, M., Melathopoulos, A.P. (2020). Label communication of acute and residual toxicity to honey bees (*Apis mellifera* L.) on insecticides of high hazard to Oregon hives does not align with federal recommendations. *Pest Management Science.* 76(5), 1664-1672.

MacPhail, V. J., S. D. Gibson, R. Hatfield, and S. R. Colla. (2020). Using Bumble Bee Watch to investigate the accuracy and perception of bumble bee (Bombus spp.) identification by community scientists. PeerJ. 8: e9412.

3.4. Website(s). List the URL for any internet site(s) that disseminates the researchactivities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

- Interactive Bee Atlas (2022): Website to display bee records to the public: https://oregonbees.osac.oregonstate.edu/
- Oregonbeeatlas.org (2021): The website contains information on the Atlas and a section for displaying findings to the public.
- PNWBumbleBeeAtlas.org (2019): Website housing all the training material and the sampling grids for volunteers to collect bumble bee records.
- PolliNation (2019): pollinationpodcast.oregonstate.edu

3.5. Have inventions, patent applications, and/or licenses resulted from the project(U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

N/A

	# Filed (Enter Numeric Value)	# Approved (EnterNumeric Value)	Patent Numbers (Separated by commas)
Patents	0	#	
Copyrights	0	#	
Trademarks	0	#	
Inventions	0	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if theymust remain confidential indefinitely. (up to 500-word limit)

Not applicable

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers,and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access andre-use these data in the future?

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation ofexisting data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

Bumble bee records from the PNW Bumble Bee Atlas have been uploaded and is publicly accessible through the website bumble bee watch: https://www.bumblebeewatch.org/

Host plant data from the Oregon Bee Atlas has been uploaded and is publicly accessible through iNaturalist: https://www.inaturalist.org/projects/ oregon-bee-atlas-plant-images-sampleid. Bee data is available here: https://oregonbees.osac.oregonstate.edu/

All bee records from the Oregon Bee Atlas are published through the Global Biodiversity Information Facility (GBIF):
Global Biodiversity Information Facility dataset of 2019 Oregon Bee Atlas (published April 2021): https://www.gbif.org/dataset/b2974853-6c41-4c63-a11b-7989e58a3ad4

Global Biodiversity Information Facility dataset of 2019 Oregon Bee Atlas (published February 2022): https://www.gbif.org/dataset/3301620d-6750-434e-97ad-abfac165bb9c#description

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4.2. If you list multiple data sets, are these data sets related? If so, please provide ashort description of how they are related.

No

4.3. Please provide copies of relevant metadata records to support FFAR's mission ofenhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

https://oregonstate.box.com/s/tmp16rducfhknxqgq66a36m3epe2mqr7

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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I willnot claim otherwise.

PI Full Name: Andony Melathopoulos

PI Signature: DocuSigned by: 5A34BD5347A44BD... Date: 9/6/2023 | 07:20:36 PDT

Authorized Signing Official (ASO) Name: Heather Toro

ASO Signature: DocuSigned by: Heather Toro 7F249ABF75694A6... Date: 9/6/2023 | 07:00:22 PDT